University of Portland Robotics

NASA Lunabotics Project Scope

2026 - 2027

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Faculty Advisor: Dr. Tribelhorn

Overview

- Team
- Project Background
- Design Challenges
- Mechanical Design
- Controls Systems
- Milestones



2026-2027 Roster

Mechanical:

- Josh Hall
- Eli Dill

Electrical:

- An Nguyen
- Diego Mendez

Comp Sci:

- Chengen Li
- Wes Helms



Lunabotics Challenge

- Contest hosted by NASA at KSC
- Teams from 40+ schools across the nation compete
- Build a robot that can operate on a simulated lunar surface
- Collect as much regolith (moon sand) as possible
- Points awarded for berm size and autonomy



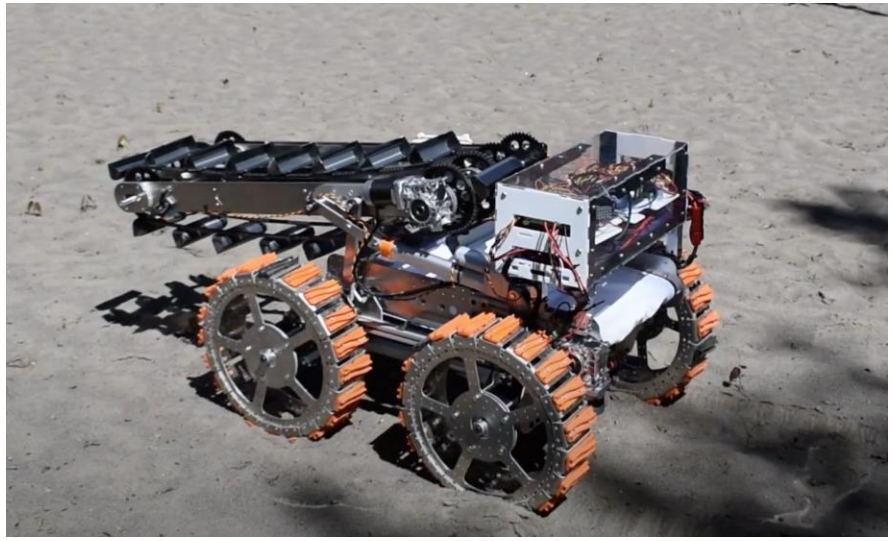
Figure 11 Artemis Arena Judges

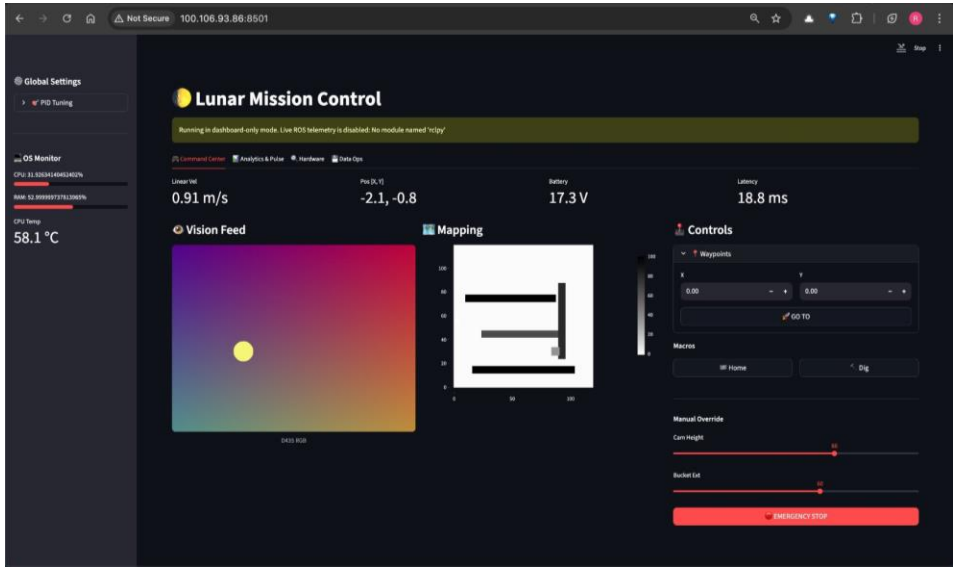


Figure 14 Mission Control Center

Last Year's Robot

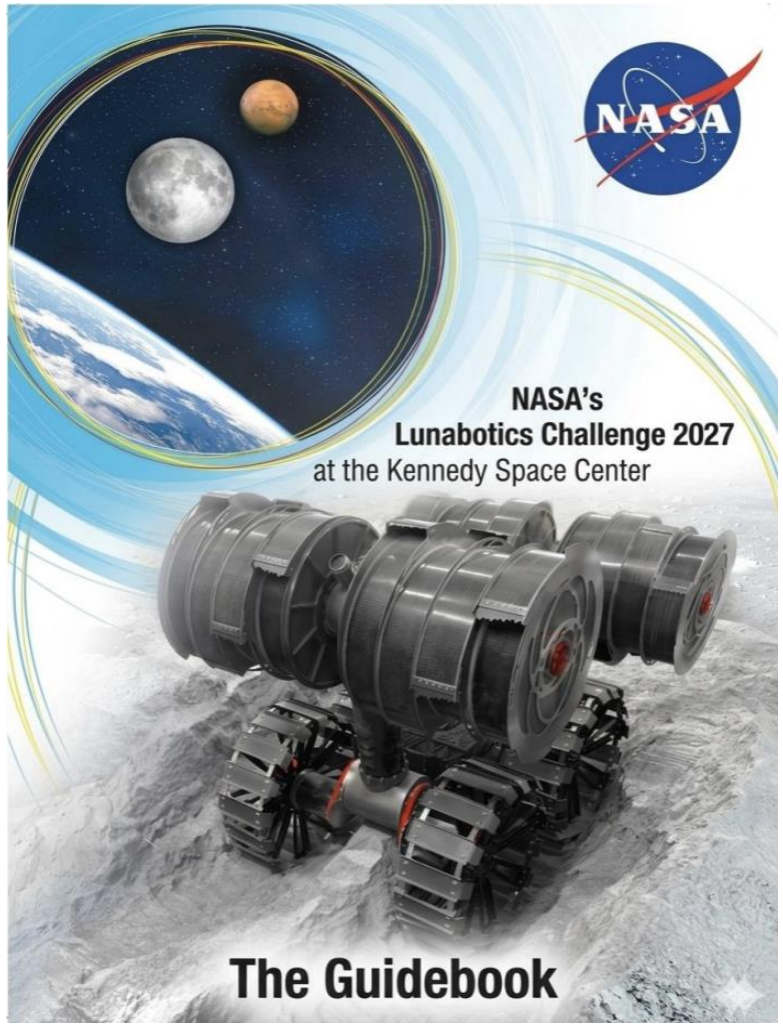
- Had good digging system and motor control
- Unable to Localize in an environment
- Bad camera latency
- Not much autonomy
- UI Dashboard features didn't work





Project Scope

- **Constraints:** guidelines set by NASA
- **Goals:**
  - Build a robot that can compete in NASA's Lunabotics Challenge with full autonomous functionality
  - Redesign the digging system



Design Challenges

Mechanical Systems

- Maximize regolith per unit weight
  - Maximize volume collected in a cycle
  - Minimize time taken to reach capacity
- Build for mechanism reliability
  - Keep heavy vibration in mind
  - Consider regolith getting in unwanted places
- Maximize repairability
  - All mechanisms can be accessed/disassembled easily

Control Systems

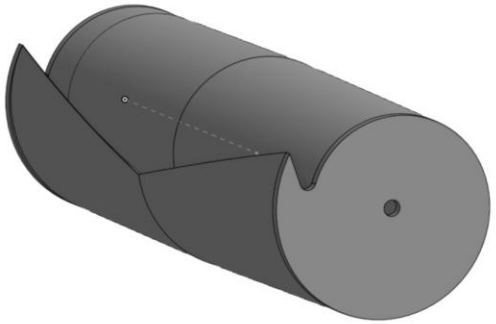
- Have to minimize bandwidth use
- No "bread crumbs" or defining waypoints via remote control during autonomous navigation
- Cannot use arena walls to localize the robot
- Organized wiring management
  - Wires don't interfere with motor systems
  - Easy to trace around each components
- E-Box Design
  - Dust-proof
  - Accessibility

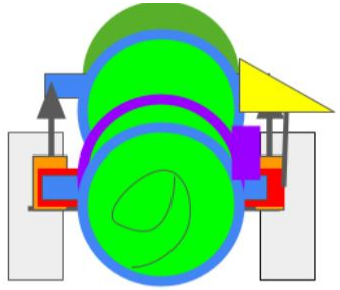
Mechanical Design

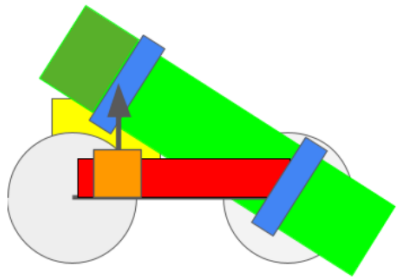
Finding a New Solution

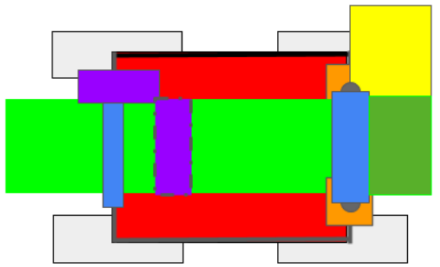
- Define the problem
  - What are we trying to do?
- Brainstorm multiple solutions
  - List all the ideas, everything is welcome
- Evaluate with quantitative properties
  - Decision matrix
- Consolidate into a solution to move forward





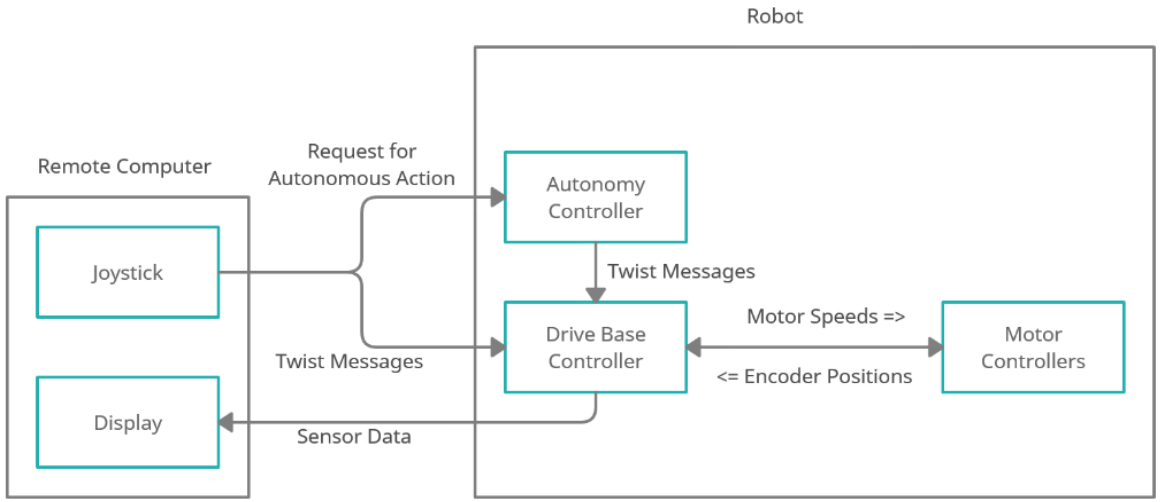




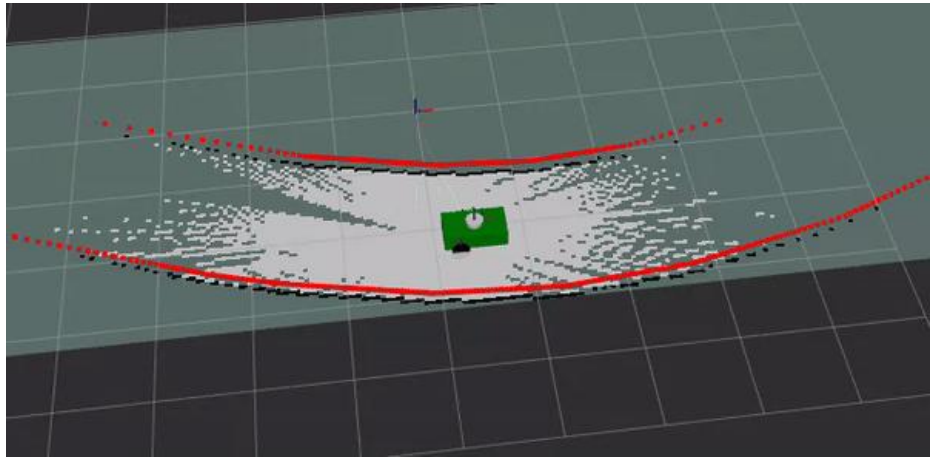


Control Systems (Software)

Revisit Old & Test New Approaches



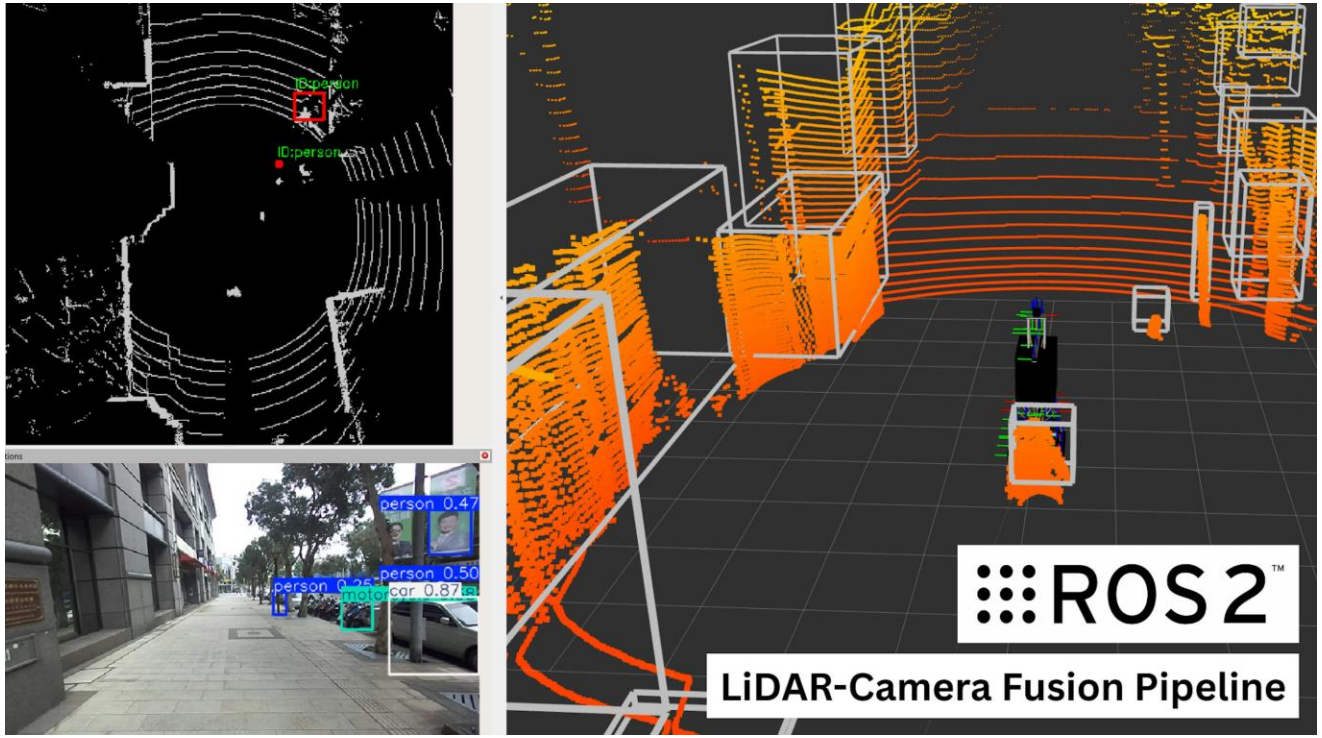
ROS2 SLAM & Nav2



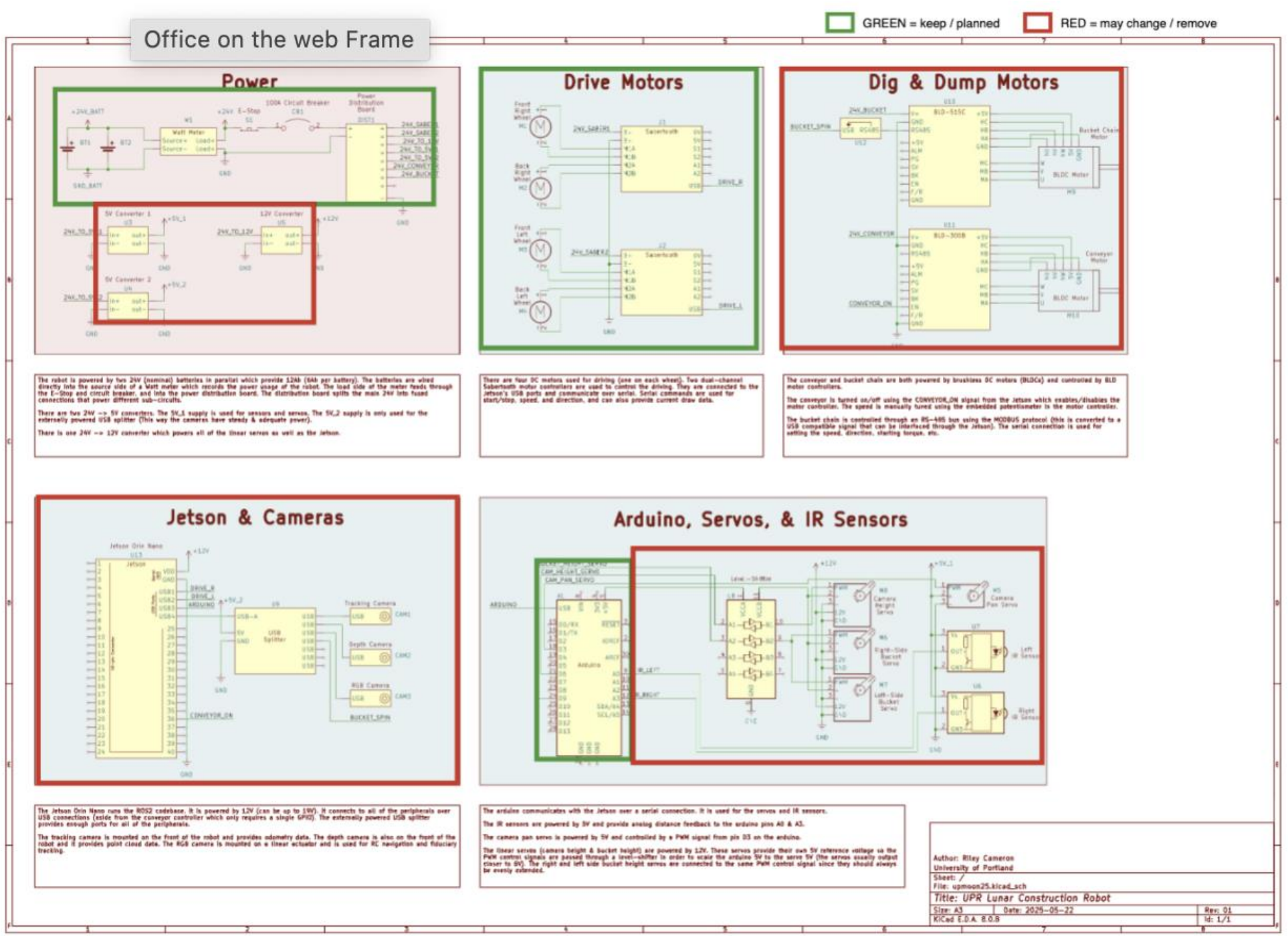
Control Systems (Software)

Test New Approaches

- Leverage new hardware
- Test existing ROS2 pipelines and packages
- Extended Kalman Filter (EKF) to perform optimal sensor fusion for mobile robot pose and velocity estimation



Current Hardware



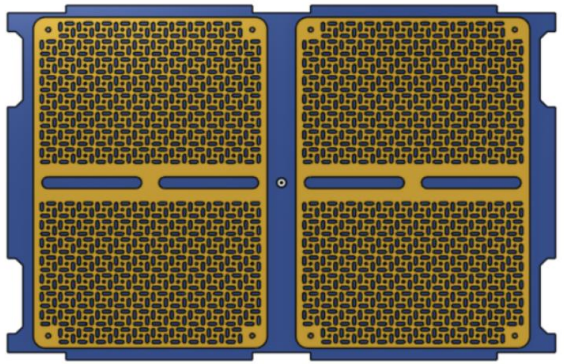
Control Systems (Hardware)

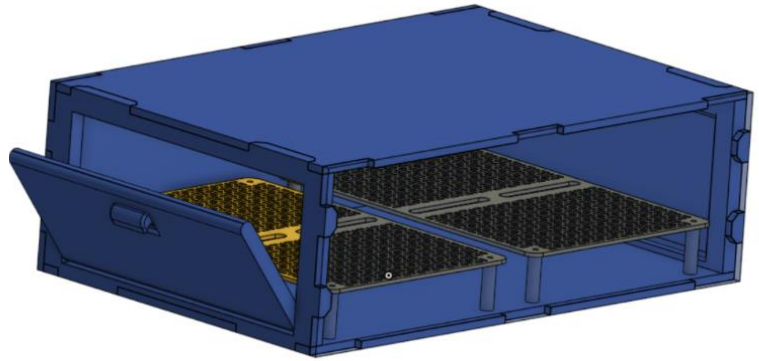
E-Box

Old E-box Design:



New E-box Design Idea:





Milestones

Fall Semester

- **Phase A**: Mission objectives, constraints, code reviews. System Requirements Review and decision matrix created
- **Phase B**: Robot architecture and autonomy approach selected. Budgets approved
- **Phase C:** Preliminary Design Review, designs, and interfaces approved
- **Phase D:** Critical subsystem prototypes demonstrate the selected design is feasible

Spring Semester

- **Phase E:** Critical Design Review, detailed design finalized for fabrication
- **Phase F:** Complete robot hardware assembled and individual subsystems bench-tested
- **Phase G:** System Integration Review, all systems operate together
- **Phase H:** Operations Readiness Review, robot passes all tests
- **Phase I:** Competition Ready!

Questions?

Image generated by Nano Banana 2

Thank you



Image generated by Nano Banana 2